

Double Integrals

The theory of integration is a bit more technical than the rest of what we've done so far, so the proofs of several theorems are omitted from Lang. It's nevertheless helpful to discuss a few of the fundamental notions involved.

A function f defined on a region R is **bounded** if for some number M , $|f(X)| < M$ for all $X \in \mathbb{R}$.

The set $[a, b] \times [c, d]$ consists of those points (x, y) such that $a \leq x \leq b$ and $c \leq y \leq d$. This is a rectangle in the plane.

A **partition** of an interval $[a, b]$ is a sequence of numbers

$$a = x_1 \leq x_2 \leq \cdots \leq x_m = b.$$

This is sometimes denoted $(x_1, x_2, \dots, x_m)$.

If we partition $[a, b]$ as $(x_1, \dots, x_m)$ and $[c, d]$ as $(y_1, \dots, y_n)$, this subdivides the rectangle $R = [a, b] \times [c, d]$ into smaller rectangles.

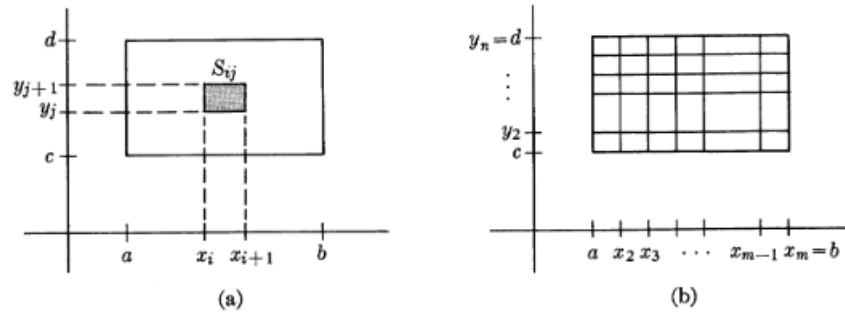


Figure 2

We denote by S_{ij} the subrectangle $[x_i, x_{i+1}] \times [y_j, y_{j+1}]$.